# Dinah J. Hartmann

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Quantitative ecologist with background and/or training in biology, conservation, computation, bioacoustics and ecological statistics. PhD student researching harbour porpoise density using passive acoustic monitoring, animal tag and visual data, computational tools and ecological statistics.

## Education

**University of St Andrews**, St Andrews, Scotland **2024-2028**

PhD in Statistics

Project: Estimating spatial and temporal patterns in density of harbour porpoises in the North Sea using datasets from passive acoustic ocean observation networks with data flow to EU Digital Twin Ocean database.

**Lund University**, Lund, Sweden **2022-2024**

Master of Science in Biology, Conservation Biology, 120 Credits, VG (distinction)

Relevant courses such as “Global Ecosystem Dynamics”, “Population and Community Ecology”, “Conservation Biology” and “Biological Monitoring”
60-credit MSc thesis on the density of harbour porpoises (*Phocoena phocoena*) using drones

**University College Dublin**, Dublin, Ireland **2018-2022**

Bachelor of Science in Environmental Biology, 240 Credits, GPA 3.81 (First-class honours)

Relevant courses such as “Ornithology”, “Animal Behaviour”, “Conservation of Biodiversity”, “Working with Biological Data”, “Population Genetics”, “Wildlife Conservation and Fisheries Management”, “Diversity of Vertebrates”, “Evolutionary Biology” and “Wildlife Habitat Modelling”

## Research

**Center for Research into Ecological and Environmental Modelling**, University of St Andrews, St Andrews, Scotland **2024-2028**

PhD thesis advised by Len Thomas, Peter Tyack and Jamie Macaulay: Processing and analysing acoustic and movement tag data in PAMGuard and MATLAB to determine the probability of detecting harbour porpoise echolocation clicks on click detectors.

**Department of Biology**, Lund University, Lund, Sweden **2023-2024**

Master of Science thesis advised by Per Carlsson and Johanna Stedt: Extensive field-work and drone-piloting experience, including focal follows of animals, programming drone paths and analysing the data in, e.g., DroneVideoMeasure, R and Excel.

Research assistance position: Supervising students in the field and independently collecting drone focal follow data of harbour porpoises.

**School of Biology and Environmental Science**, University College Dublin, Dublin, Ireland **2022**

Bachelor of Science thesis advised by Tancredi Caruso on the effects of drought on soil biodiversity: Ecological laboratory work to determine soil biodiversity, extracting and preserving biota, analysing them under microscopes.

## Teaching and Outreach

**School of Mathematics and Statistics**, University of St Andrews, St Andrews, Scotland **2025-2028**

School Seminar Series organisation (Service role) **2025-2026**

MT2508 Statistical Inference: Lead demonstrator for statistical programming in R, provided R statistics and programming support to students **2026**

MT2508 Statistical Inference: Lead demonstrator for statistical programming in R, provided R statistics and programming support to students **2025**

MT2508 Statistical Inference: Tutor in statistics, Prepared and held small group tutorials, marked assignments and gave feedback **2025**

**School of Biology**, University of St Andrews, St Andrews, Scotland **2025-2028**

Women in Ocean Science event presenter **2026**

Volunteer at the European Cetacean Society Conference (organise/facilitate) **2026**

Women in Ocean Science event presenter **2025**

**Department of Biology**, Lund University, Lund, Sweden **2023-2024**

Teaching drone methodology (piloting, safety, marine mammal focal follows and flight path programming) to Master's students

TV-documentary segment on harbour porpoise drone research on ARTE ("Night Trains – Von Berlin nach Stockholm")

Feature in 'Sveriges Natur', the biggest Swedish nature magazine ("Kampen för att rädda tumlaren")

Smaller newspaper articles in 'tidningensyre' ("Snärjs i fisknät – kan tumlarna räddas från dödsdansen?" and 'Helsingborg dagblad' ("Barndomens kärlek till Sveriges enda val blev hennes karriär")

**Seal Rehabilitation Norden-Norddeich**, Norden, Germany **2021**

Voluntary work in environmental education and outreach: Insights into seal conservation and monitoring, public outreach talks, teaching museum visitors in seal conservancy work via small talks and personal interactions

## Publications

Hartmann, D.J., J.D.J. Macaulay, L. Rojano-Doñate, J. Teilmann, S.L. Elmegaard, P.T. Madsen, U. Siebert, P. Tyack and L. Thomas. (2026), Acoustic detection probability of harbour porpoises depends on behaviour, detection unit and click-train classification. JASA.

Hartmann, D., Palmqvist, V. and Stedt, J. (2025), Programmed unmanned aerial vehicles show great potential for monitoring marine megafauna in specific areas of interest. Remote Sens Ecol Conserv. https://doi.org/10.1002/rse2.70043

## Workshops

**Data Flow into the Digital Twin Ocean: Integration of C/F-POD PAM Data.** Ellen Jacobs, Elisabeth Debusschere, Mia Kronberg-Nielsen, Dinah Hartmann and Emily Griffiths**.** Society for Marine Mammalogy (SMM) 2026 Conference, Puerto Rico. (October 2026).

## Talks and Posters

**Hartmann, D.**, Macaulay, J. D. J., Tyack, P., Rojano-Doñate, L., Teglberg-Madsen, P., Teilmann, J., Siebert, U. & Thomas, L.

[Talk]. Society for Marine Mammology Conference, San Juan, Puerto Rico. Harbour porpoise behavioural classification using acoustic detections for density estimation. (October 2026).

**Hartmann, D.**, Macaulay, J. D. J., Tyack, P., Rojano-Doñate, L., Teglberg-Madsen, P., Teilmann, J., Siebert, U. & Thomas, L. The probability of detecting harbour porpoise echolocation clicks.

[Poster]. European Cetacean Society Conference, Dundee, Scotland. (April 2026).

[Invited Seminar]. Aarhus University Ecoscience Group, Roskilde, Denmark. (January 2026).

[Conference talk]. Marine Alliance for Science and Technology Scotland, Glasgow, Scotland. (November 2025).

[Seminar]. Scottish Oceans Institute, St Adnrews, Scotland. (October 2025).

[Poster]. International Bioacoustics Congress, Kerteminde, Denmark. (September 2025).

[Invited Talk]. Aarhus University Bioacoustics Group, Aarhus, Denmark. (September 2025).

**Hartmann, D.**, Palmqvist, V. & Stedt, J. En route to standardisation: Porpoise monitoring using a programmed UAV.

[Poster]. European Cetacean Society 36th conference, Ponta Delgada, Azores, Portugal. (May 2025).

**Hartmann, D.**, Macaulay, J. D. J., Tyack, P., & Thomas, L. Harbour porpoise click detection.

[Poster]. UK and Ireland regional student chapter for marine mammology student conference, St Andrews, Scotland. (January 2025).

**Hartmann, D.**, Stedt, J., & Carlsson, P. Spatio-temporal distribution of harbour porpoises around Kullaberg nature reserve.

[Invited Talk]. Danish Marine Mammal Society Symposium, Denmark. (March 2024).

## Awards, Grants, Scholarships

**Doctoral studentship**, EU funded as part of the DTO-Bioflow project (~120,000 gbp) **2024-2028**

**Environmental Biology Medal** for highest overall Bachelor's GPA **2022**

**Environmental Biology Stage 4 Scholarship** for highest class GPA (1,000 euro) **2021**

## Skills and Trainings

**Scottish Mathematical Sciences Training Centre** **2024-2025**

Modern Regression and Simulation methods and Bayesian statistics

**University of Oxford**, Oxford, England **December 2024**

Academy for PhD training in statistics (Statistical Inference and computing)

**University of New Hampshire**, Durham, USA **June 2024**

SeaBASS summer school in bioacoustics

**Computing, quantitative**

R

RStan

MATLAB

ArcGIS Pro

QGIS

PAMGuard

Raven

SPSS

**Other research skills**

A1-A3 Drone License, drone piloting and programming

LaTeX/Overleaf

Microsoft office applications

German (native proficiency)

French

Spanish

Laboratory skills (soil drying, extraction of biota, microscopes, DNA extraction)